Data visualization

Data Science Flipped Classroom - Module 2

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ggplot2

What is ggplot2?

ggplot2 is an R package for data visualization based on the **Grammar of Graphics**.

It lets you build plots by combining components in a structured way, like building sentences.

Grammar of Graphics

Example:

> "Make a bar chart of sales by region and products."

In a structured way, translates to:

> "Using the sales data, put region on the x-axis, sales on the y-axis, and group bars by product with a unique color for each region."

Each part of the sentence maps to a component of the plot.

Essential building blocks

The three essentials

1. **Data:** a `data.frame` or `tibble`, passed to `ggplot()`
2. **Aesthetics:** how variables map to visual properties (position, color, size), defined inside `aes()`
3. **Layers:** where data takes shape (points, lines, bars), added with `geom_*()` or `stat_*()` functions

> **Good to know**
>
> With just these three elements you can already generate a plot!

The mpg dataset

mpg is a dataset included in ggplot2 with fuel economy data for 234 cars from 1999 to 2008.

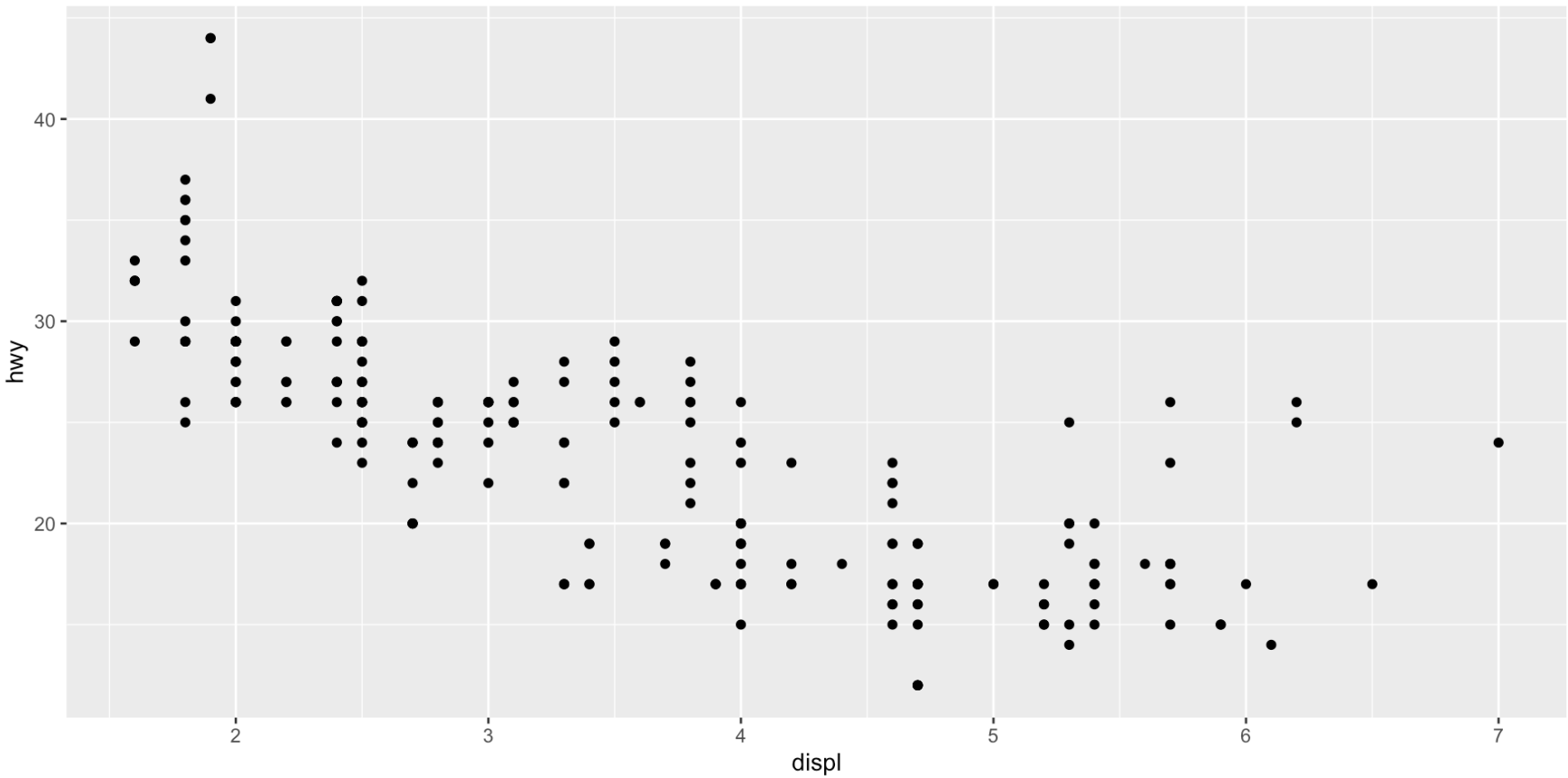
```r
mpg |>
  select(manufacturer, model, displ, year, cyl, hwy) |>
  head(5)
```

```
# A tibble: 5 × 6
  manufacturer model displ  year   cyl   hwy
  <chr>        <chr> <dbl> <int> <int> <int>
1 audi         a4      1.8  1999     4    29
2 audi         a4      1.8  1999     4    29
3 audi         a4      2    2008     4    31
4 audi         a4      2    2008     4    30
5 audi         a4      2.8  1999     6    26
```

Example: a basic plot

```r
ggplot(data = mpg, aes(x = displ, y = hwy)) +
  geom_point()
```

Example: a basic plot



Breaking it down

```r
ggplot(data = mpg, aes(x = displ, y = hwy)) +
  geom_point()
```

Component	Code	Role
Data	`data = mpg`	The dataset
Aesthetics	`aes(x = displ, y = hwy)`	Map variables to axes
Layer	`geom_point()`	Draw points

Additional components

Scales

Scales fine-tune how data is visually represented.

With `scale_*()` functions you can:

- Change color palettes and shapes
- Modify axis labels and limits
- Control legend behavior

Coordinates

Coordinates define the spatial context of the plot.

With `coord_*()` functions you can:

- Apply spatial transformations (e.g., `coord_flip()`)
- Set aspect ratios
- Zoom into a region of the plot

Facets

Facets split a plot into multiple panels based on subsets of the data.

- `facet_wrap()`: panels from one variable
- `facet_grid()`: panels from two variable combinations

Useful to uncover patterns across groups!

Themes

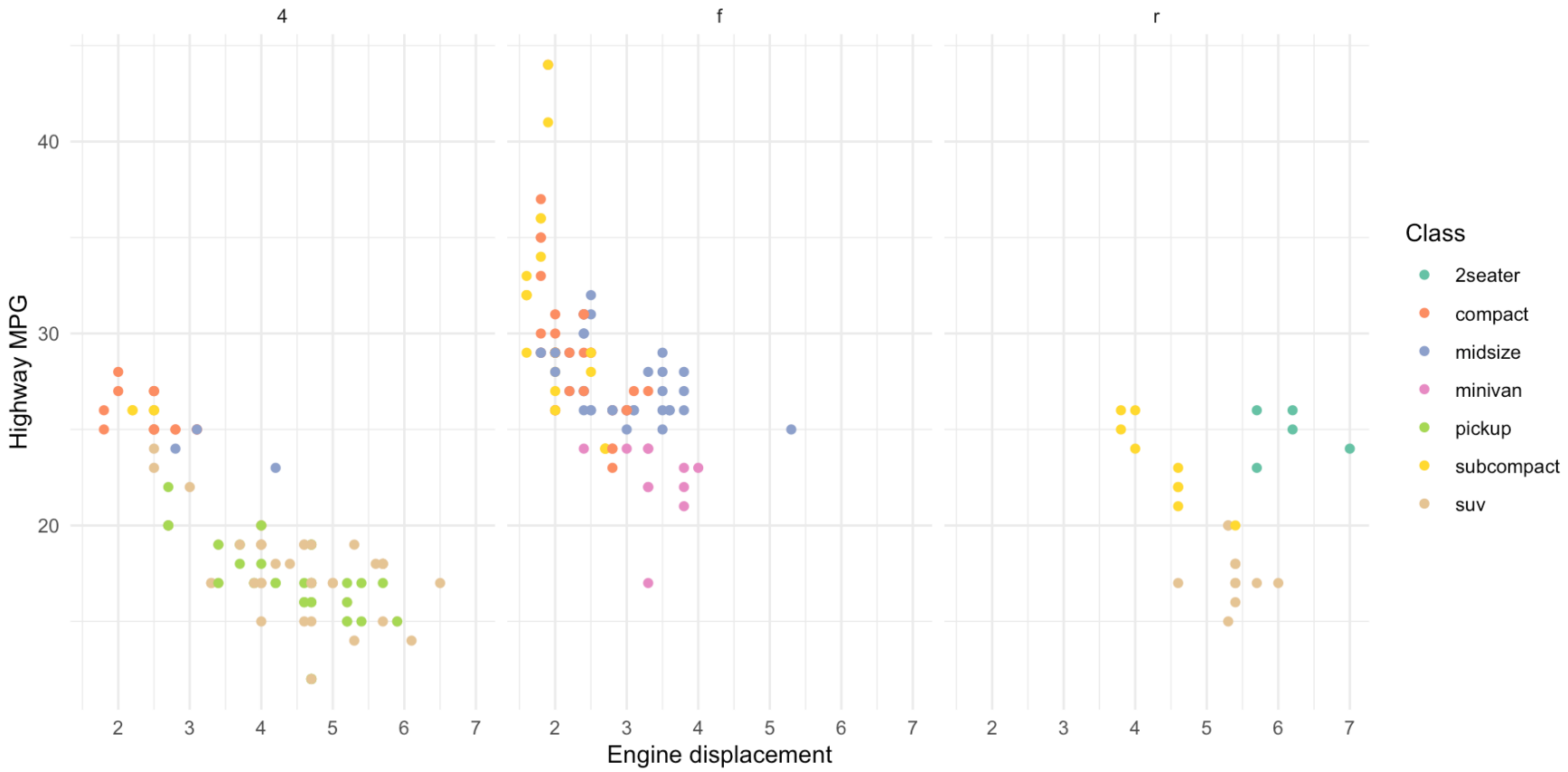
Themes control the overall visual style of the plot.

- `theme_*()` functions apply a complete style (e.g., `theme_minimal()`)
- `theme()` lets you fine-tune individual elements (text, backgrounds, gridlines)

The full “sentence”

```r
ggplot(data = mpg, aes(x = displ, y = hwy, color = class)) +
  geom_point() +
  scale_x_continuous(name = "Engine displacement") +
  scale_y_continuous(name = "Highway MPG") +
  scale_color_brewer(name = "Class", palette = "Set2") +
  facet_wrap(~drv) +
  theme_minimal()
```

The full “sentence”



All the components together

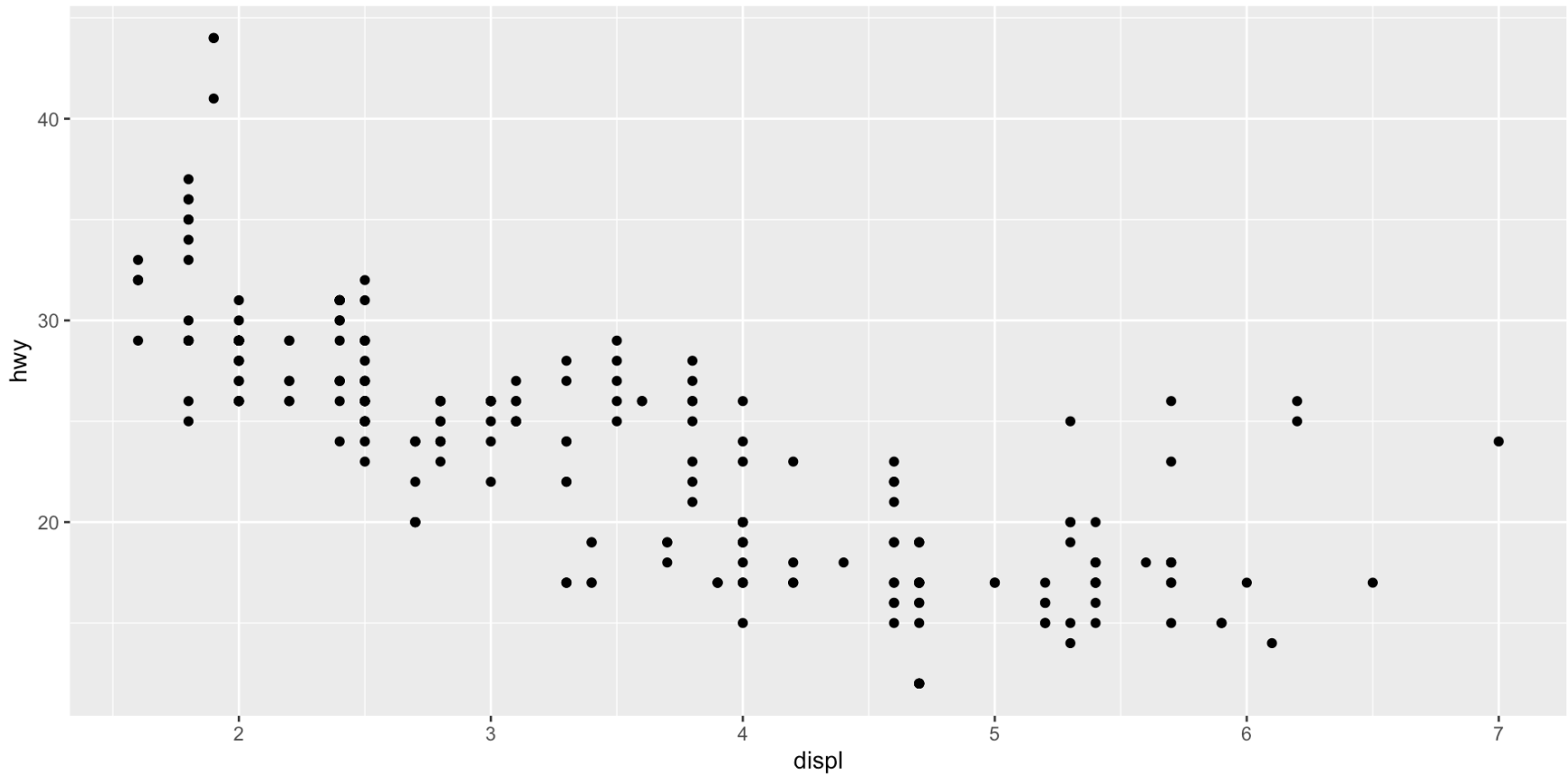
Component	Function	Purpose
Data	`ggplot(data)`	The dataset to visualize
Aesthetics	`aes()`	Map variables to visual properties
Layers	`geom_*()` / `stat_*()`	Geometric shapes or summaries
Scales	`scale_*()`	Fine-tune visual mappings
Coordinates	`coord_*()`	Spatial transformations
Facets	`facet_*()`	Split into panels
Themes	`theme_*()` / `theme()`	Visual style

Common plot types

Scatter plot: `geom_point()`

```
ggplot(mpg, aes(x = displ, y = hwy)) +
  geom_point()
```

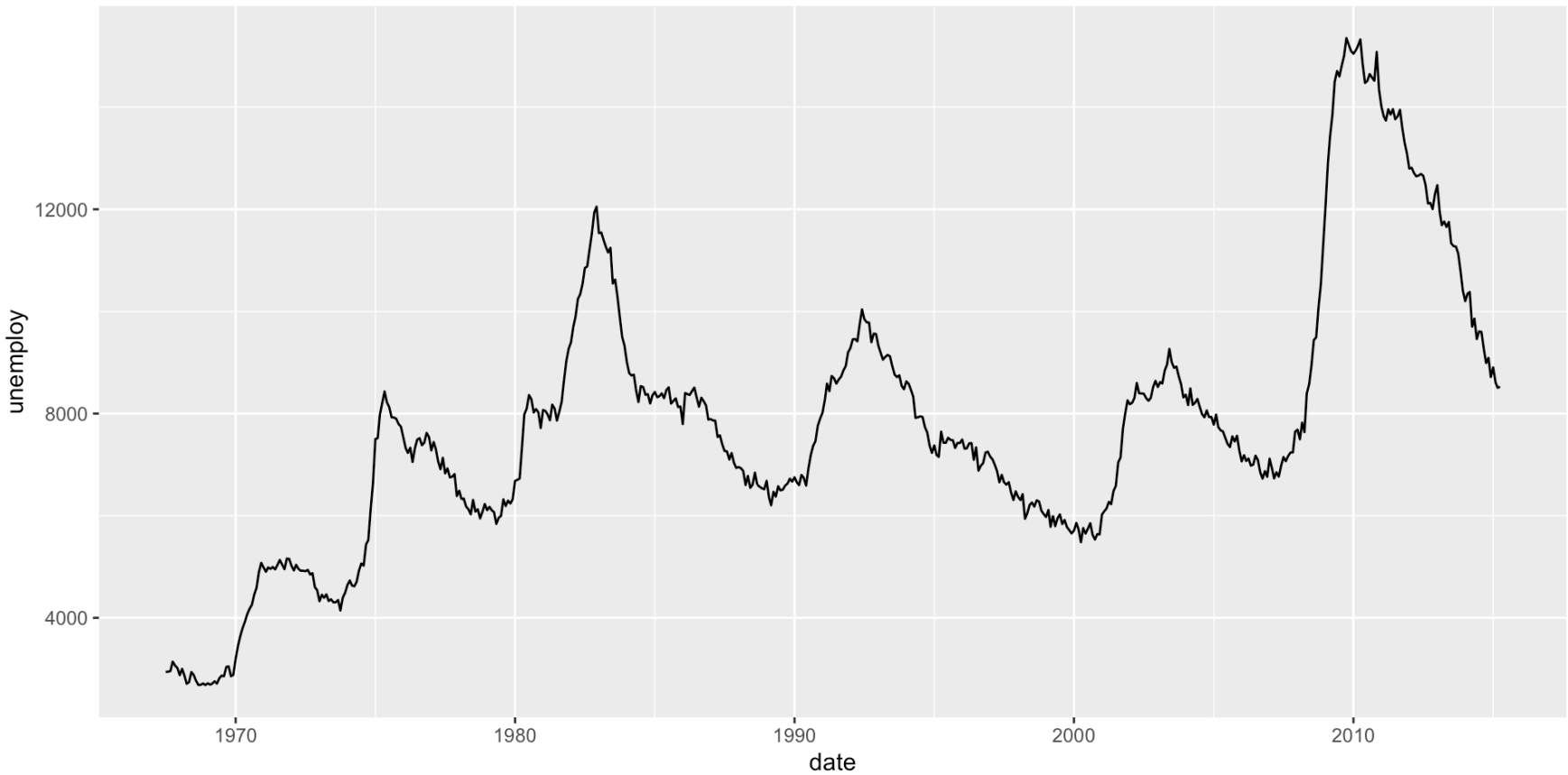
Scatter plot: `geom_point()`



Line plot: `geom_line()`

```
ggplot(economics, aes(x = date, y = unemploy)) +
  geom_line()
```

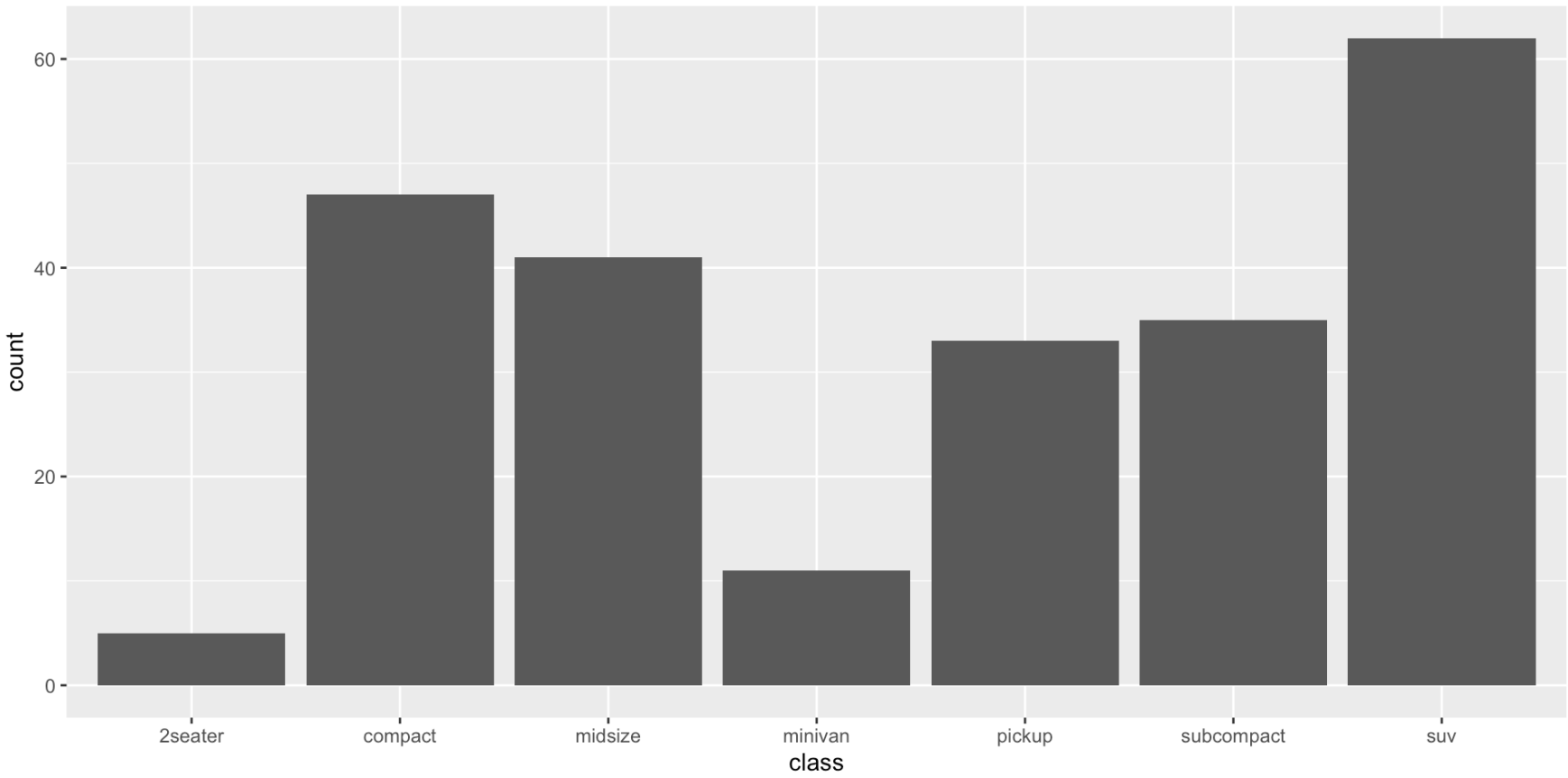
Line plot: `geom_line()`



Bar chart: `geom_bar()`

```
ggplot(mpg, aes(x = class)) +
  geom_bar()
```

Bar chart: `geom_bar()`

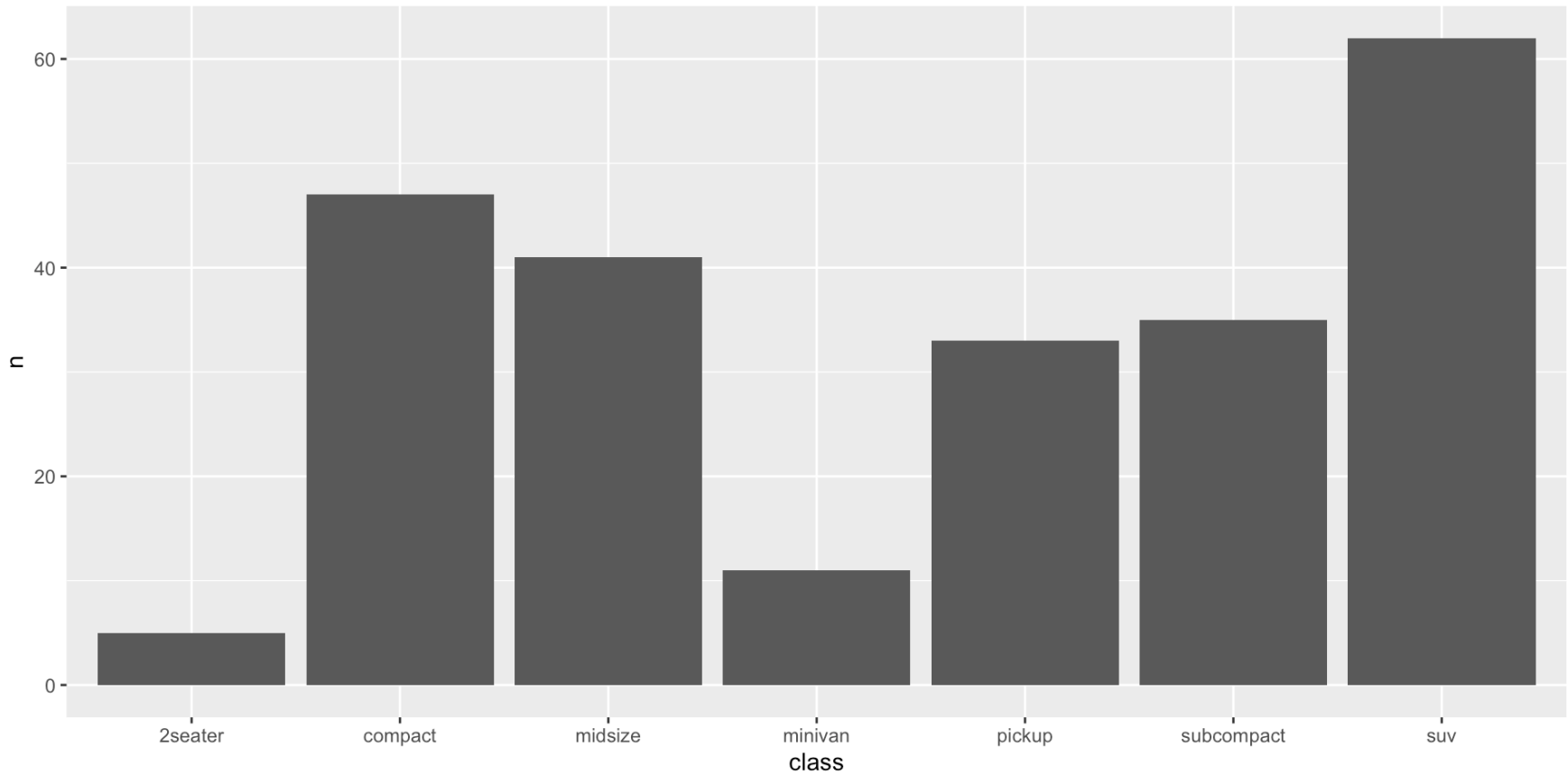


Column chart: `geom_col()`

```
mpg |>
  count(class) |>
  ggplot(aes(x = class, y = n)) +
  geom_col()
```

> `geom_bar()` counts for you, `geom_col()` uses values you provide.

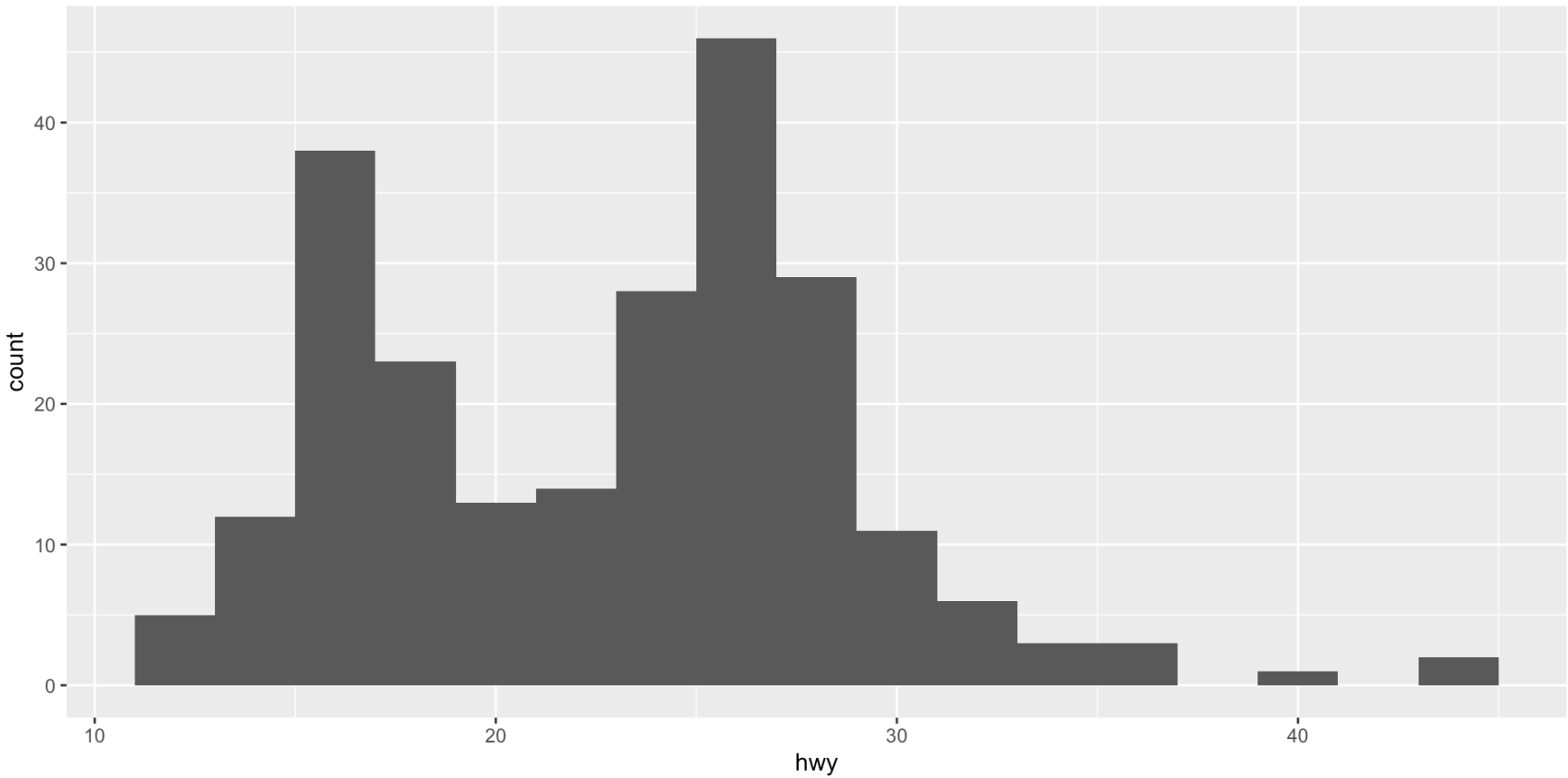
Column chart: `geom_col()`



Histogram: `geom_histogram()`

```r
ggplot(mpg, aes(x = hwy)) +
  geom_histogram(binwidth = 2)
```

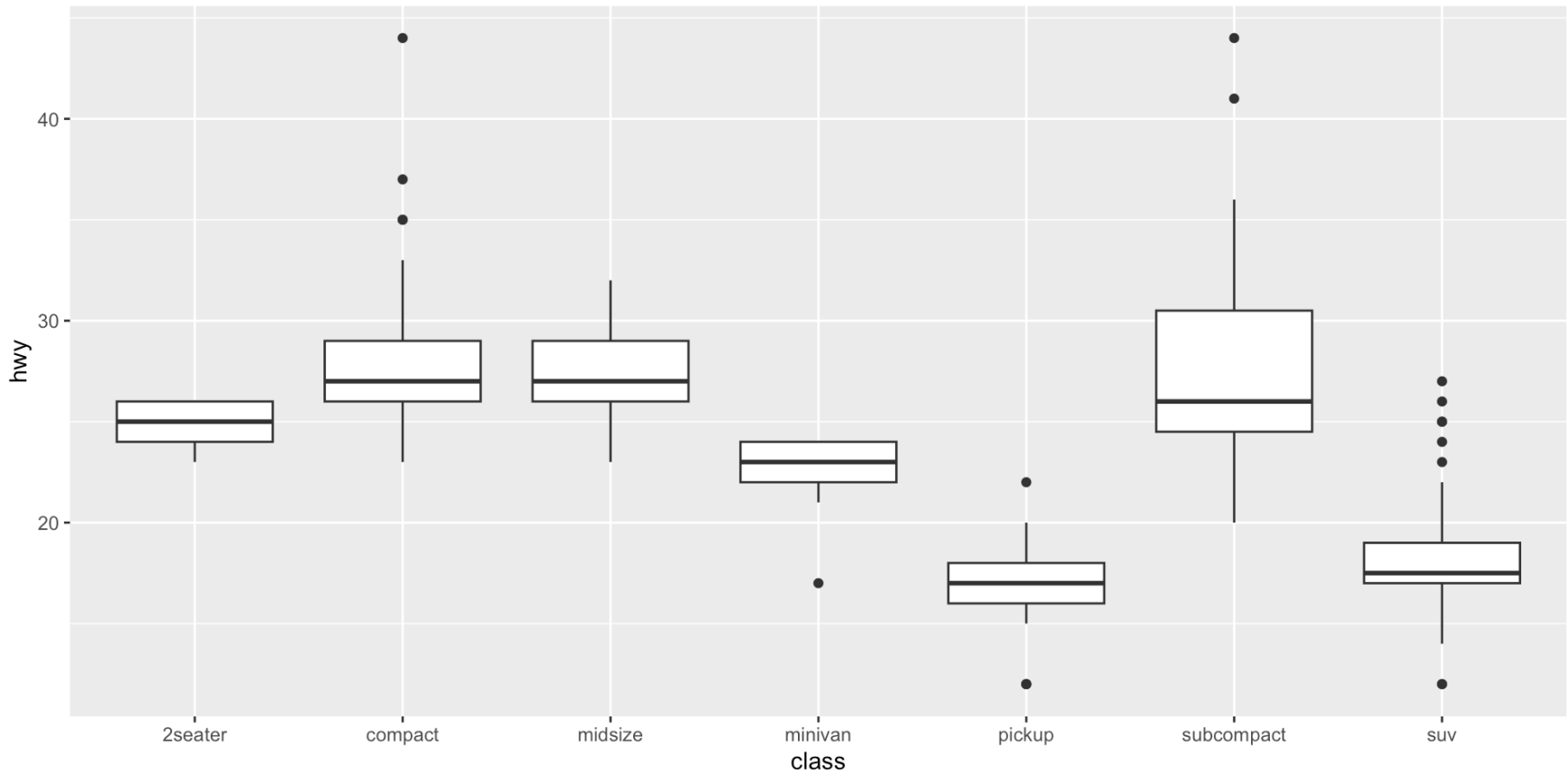
Histogram: `geom_histogram()`



Box plot: `geom_boxplot()`

```
ggplot(mpg, aes(x = class, y = hwy)) +
  geom_boxplot()
```

Box plot: `geom_boxplot()`

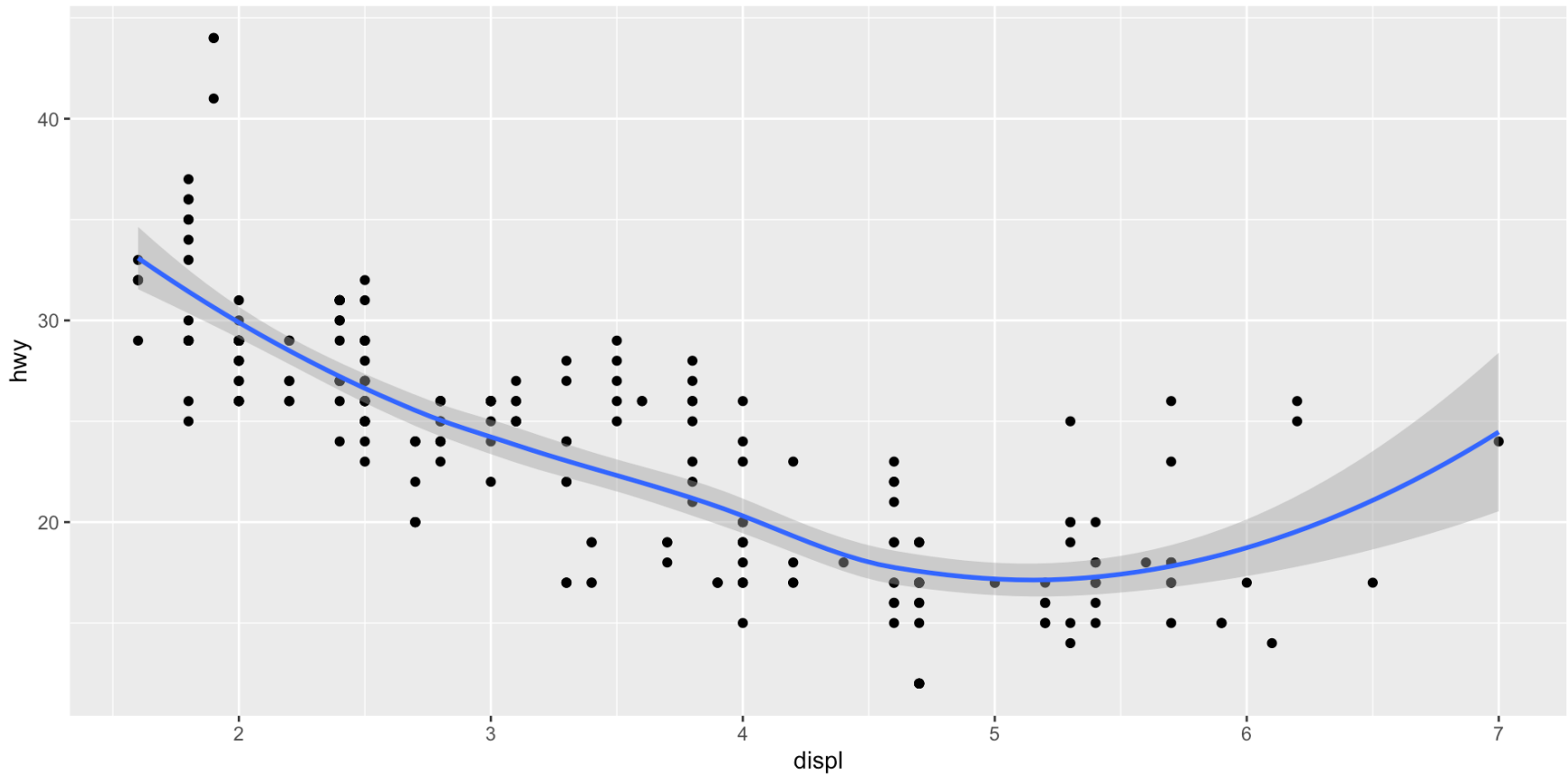


Smooth line: `geom_smooth()`

```
ggplot(mpg, aes(x = displ, y = hwy)) +
  geom_point() +
  geom_smooth()
```

> You can layer multiple `geom_*()` functions on the same plot!

Smooth line: `geom_smooth()`

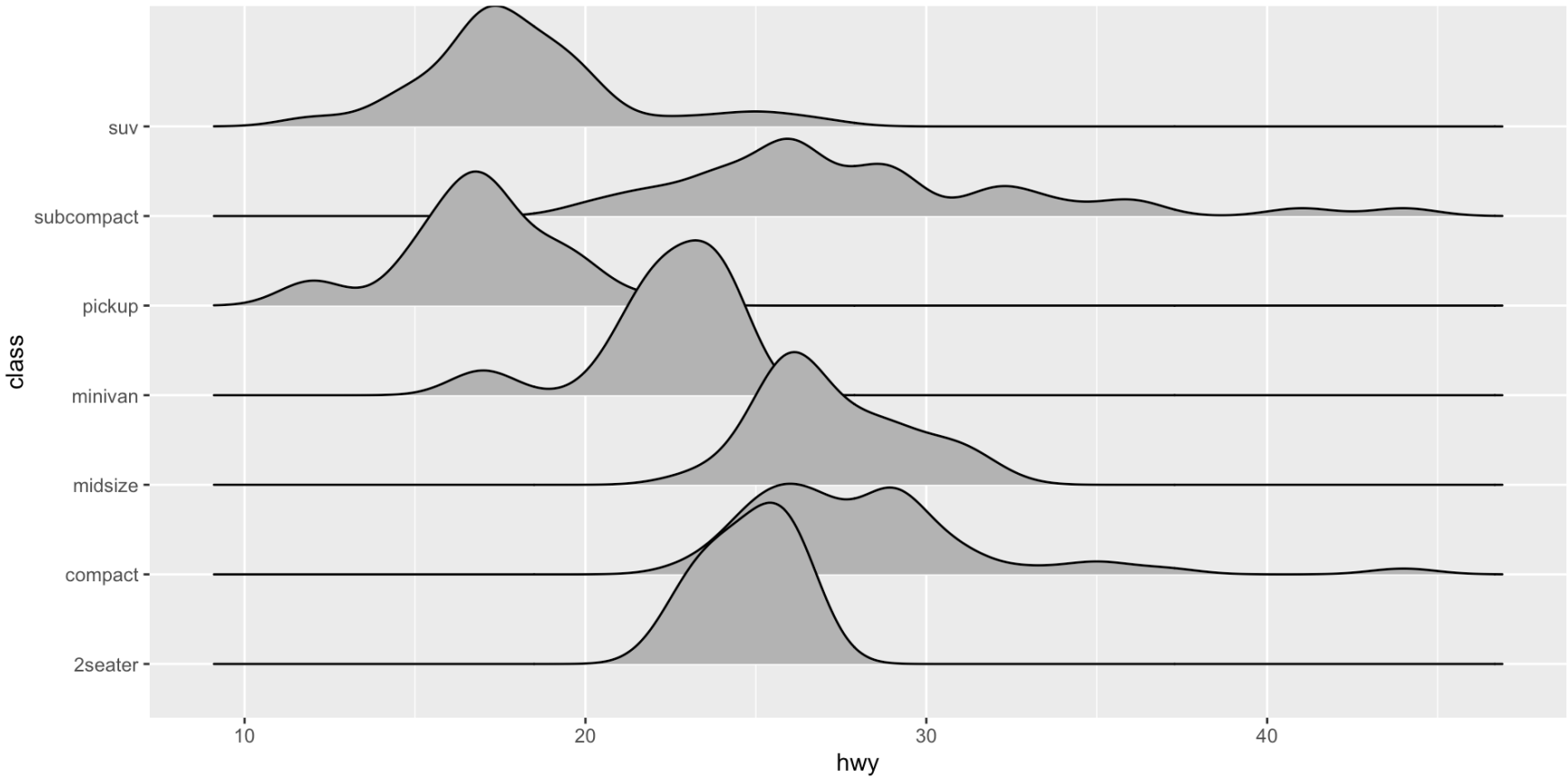


Ridge plot: `geom_density_ridges()`

```r
ggplot(mpg, aes(x = hwy, y = class)) +
  geom_density_ridges()
```

> Ridge plots are great for comparing distributions across groups. Requires the `ggridges` package.

Ridge plot: `geom_density_ridges()`

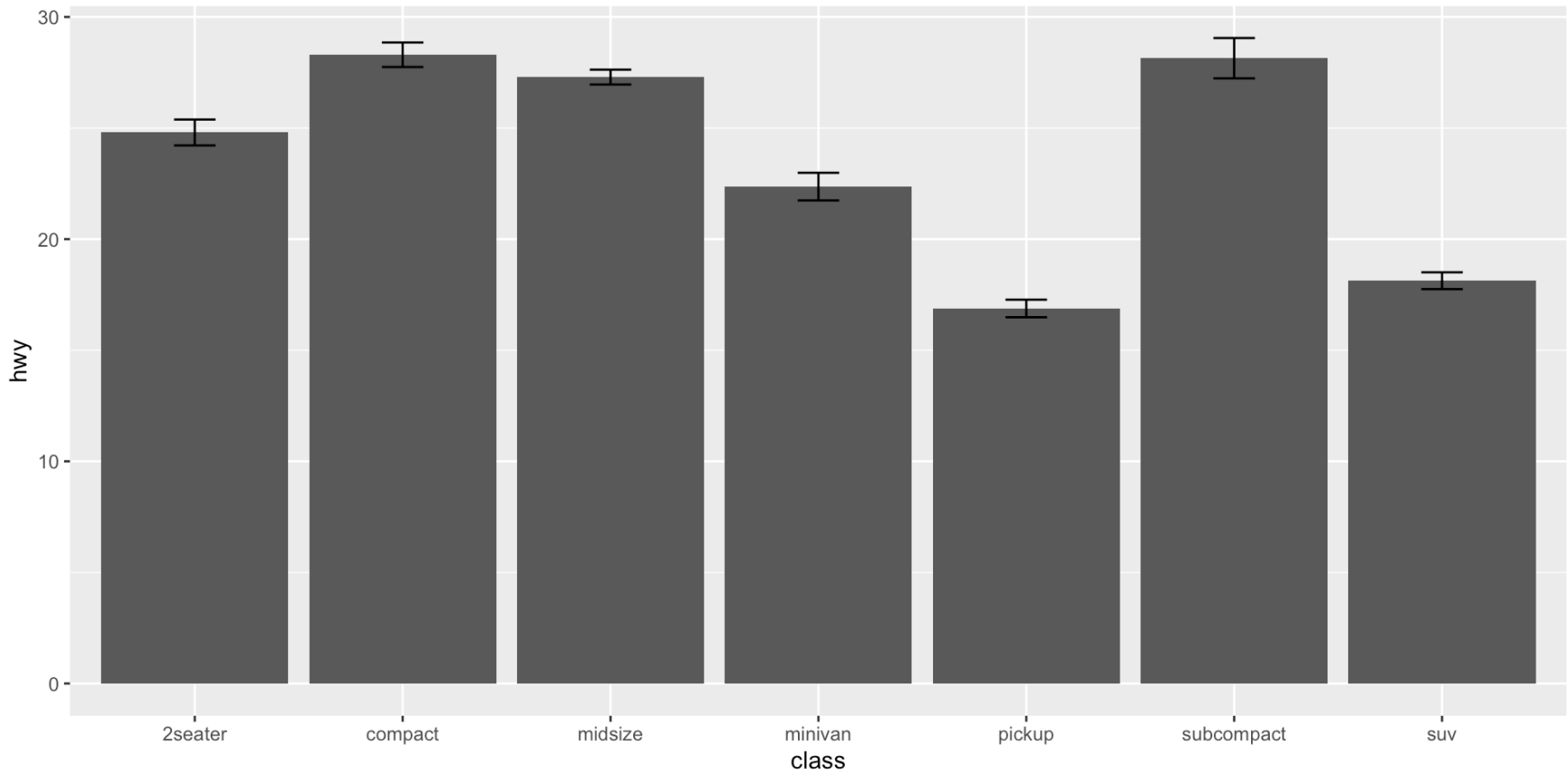


Summary with `stat_summary()`

```
ggplot(mpg, aes(x = class, y = hwy)) +
  stat_summary(fun = mean, geom = "col") +
  stat_summary(fun.data = mean_se, geom = "errorbar", width = 0.2
```

> `stat_*()` functions compute summaries before plotting.

Summary with `stat_summary()`



Choosing the right plot

What plot should I use?

It depends on the type of variable you want to visualize.

Types of variables

Non-numerical

- **Categorical**: no inherent order (e.g., color, country)
- **Ordinal**: with a meaningful order (e.g., education level, rating)

Numerical

- **Discrete**: countable values (e.g., number of children)
- **Continuous**: any value in a range (e.g., temperature, weight)

Can you tell the type?

Does it make sense to do math with this variable?

Zip code: 10115, 28001, 75008...

Looks like a number, but...

What is the **average zip code**?

It makes no sense. Zip codes are categorical.

Year: 2019, 2020, 2021...

It depends on the context!

- Comparing groups (sales in 2020 vs 2021): categorical
- Tracking change over time: continuous

> The same variable can be treated differently depending on your question.

Likert scale: 1, 2, 3, 4, 5

A satisfaction survey asks: “Rate from 1 (very bad) to 5 (excellent).”

Is the difference between 1 and 2 the same as between 4 and 5?

Not necessarily. These are ordinal, not truly numerical.

Age group: “18-25”, “26-35”, “36-45”...

The values contain numbers, but...

You can’t average “18-25” and “26-35”.

These are ordinal: categories with a meaningful order.

“How many hours did you sleep?” 4, 6, 7, 8...

Looks continuous, but people usually answer in whole numbers.

Is “7 hours” really 7.0, or just a rounded estimate?

In a survey, this is arguably discrete: the responses are countable, even if the underlying variable is continuous.

Matching variables to plots

Variables	Plot type	`geom_*()`
1 categorical	Bar chart	`geom_bar()`
1 continuous	Histogram	`geom_histogram()`
1 continuous	Density	`geom_density()`
2 continuous	Scatter plot	`geom_point()`
2 continuous	Line plot	`geom_line()`
1 categorical + 1 continuous	Box plot	`geom_boxplot()`
1 categorical + 1 continuous	Ridge plot	`geom_density_ridges()`
1 categorical + 1 continuous	Column chart	`geom_col()`
2 categorical	Stacked/grouped bar	`geom_bar()`

Questions?